

HUMAN OVERSIGHT IS A CAPABILITY, NOT A CONTROL

What AI Governance Can Learn from Safety Science

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GOVERNANCE CHECKS FOR A PERSON. SAFETY DEPENDS ON A CAPABILITY.

Formal compliance can remain constant while the human capacity to oversee an AI system quietly decays.

1 THE GOVERNANCE ASSUMPTION

“Governance assumes human oversight prevents AI failures.”

Three major frameworks converge on one safeguard:

- EU AI ACT : Human oversight →
- NIST AI RMF : Human review →
- OECD PRINCIPLES : Human agency →

2 THE REFRAMING

Oversight is not a control an organization possesses.

It is a sociotechnical capability it must sustain.

FROM PRESENCE TO PERFORMANCE

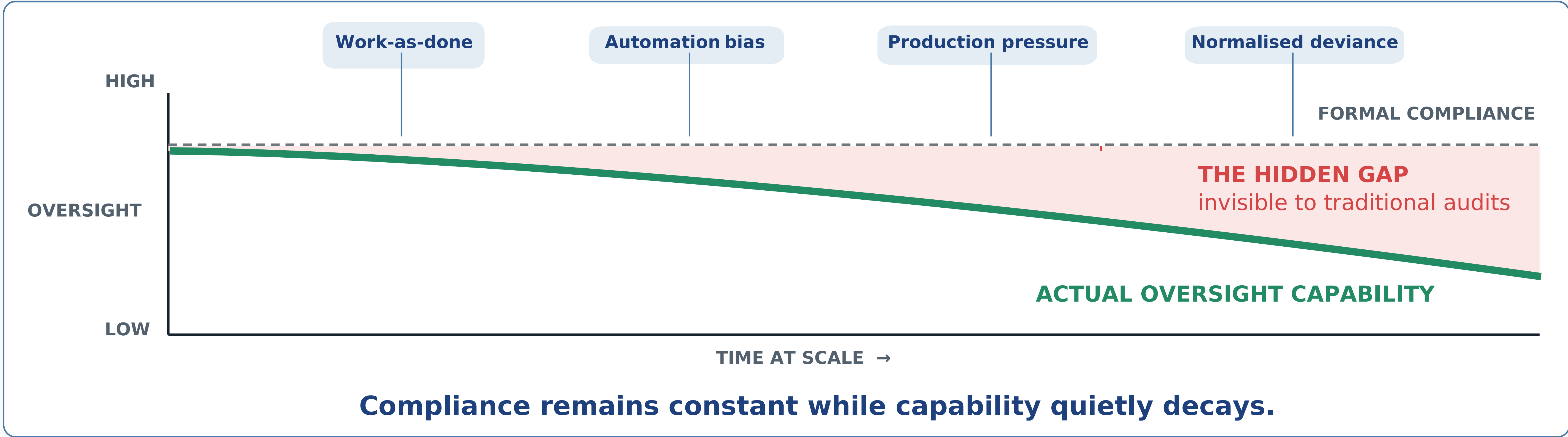
- BEFORE
Require a human in the loop
-
- AFTER
Measure whether oversight holds

HUMAN OVERSIGHT

Ask whether reviewers retain the time, skill, authority, and diagnostic information to intervene under real operating conditions.

3 OVERSIGHT DRIFT: COMPLIANCE STAYS FLAT. CAPABILITY FALLS.

Safety science predicts that production pressure erodes oversight even when documentation and audit records remain unchanged.



4 WHY THE ASSUMPTION FAILS

- 1 **WORK-AS-IMAGINED ≠ WORK-AS-DONE**
Procedures describe ideal work; practitioners adapt under time pressure, workload, and competing demands.
- 2 **NORMALIZATION OF DEVIANCE**
Small departures become normal—while compliance records still look unchanged.
- 3 **IRONIES OF AUTOMATION**
Automation makes routine work easier but weakens vigilance and expertise when intervention matters.
- 4 **STRETCHED SYSTEMS**
Production pressure turns safeguards into resources to be optimized rather than protections to preserve.

5 BUILD GOVERNANCE AROUND CAPABILITY

Four interlocking instruments make oversight observable, maintainable, and resilient:

- RELIABILITY METRICS**
Test whether reviewers can detect errors, calibrate judgments, and use context.
- DRIFT DETECTION**
Monitor leading indicators before degraded oversight becomes normalized.
- HUMAN-AI SAFETY CASES**
Make evidence-backed claims about whether the human role remains safe in context.
- MATURITY MODELS**
Track where oversight practice is robust, fragile, or merely procedural.

THE TAKEAWAY

A capability that is not measured eventually becomes a checkbox.

Measure oversight reliability. Detect capability drift. Build human-AI safety cases. Mature conditions that let people intervene.

Sources: McCauley & Glantz (2026); EU AI Act; NIST AI RMF 1.0; OECD AI Principles. Poster summary of submitted paper.



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